Study ID	SID1	**Paper Title**	GoodSpread: Criticality-Aware Static Scheduling of CPS with Multi-QoS Resources							
Publication Year	2020	**Source**	IEEE	**Authors**	Debayan Roy; Sumana Ghosh; Qi Zhu; Marco Caccamo; Samarjit Chakraborty					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 11898-2:2016 (CAN); AUTOSAR Classic Platform Standards									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Environment, Comms/network, Software/timing									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Worst-case bounds	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Architectural techniques, Assurance/validation, Optimization	**Technique Name**	GoodSpread							
Inputs	Plant/FlexRay	**Outputs**	Schedules							
Mitigation Action	Spread high-critical slots offline before deployment.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Control apps					
Reproducibility	Yes	No	Partial	**Metrics Reported**	QoS usage, runtime, settling time, extensibility					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	1	1	1	1	0.5	0.5
Component Focus	Scheduling	**Safety-Critical scenario**	FlexRay	**System Type**	System-level					
Contribution of the paper	Introduces spread-aware static multi-QoS CPS scheduling.									
Summary	GoodSpread statically schedules scarce high-QoS CPS resources by spreading critical slots over time to tolerate uncertain disturbance/criticality timing, while maximizing schedule extensibility.									
Any Other Comments										

Study ID	SID2	**Paper Title**	A Feature-Based Machine Learning Approach for Mixed-Criticality Systems							
Publication Year	2021	**Source**	IEEE	**Authors**	Nelson Vithayathil Varghese; Akramul Azim; Qusay H. Mahmoud					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 26262, IEC 61508									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	ML/data-driven, Environment									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation; Ranking	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Architectural techniques, Design-time modeling, Assurance/validation	**Technique Name**	-							
Inputs	Feature space	**Outputs**	Partitions							
Mitigation Action	Route weak partitions to regular ML.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	MULAN scene					
Reproducibility	Yes	No	Partial	**Metrics Reported**	Hamming loss, partition rank					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	0.5	0.5	0.5	0.5	0	0.5
Component Focus	Partitioning	**Safety-Critical scenario**	Generic	**System Type**	Component-level					
Contribution of the paper	Routes high-confidence feature partitions to Safe-ML.									
Summary	Proposes feature-space partitioning for ML in mixed-criticality CPS: high-confidence partitions go to Safe ML, weaker partitions go to regular ML or rejection.									
Any Other Comments										

<table>
<tr><td>Study ID</td><td>SID3</td><td>Paper Title</td><td colspan="6">Using Rigorous Simulation to Support ISO 26262 Hazard Analysis and Risk Assessment</td></tr>
<tr><td>Publication Year</td><td>2015</td><td>Source</td><td>IEEE</td><td>Authors</td><td colspan="4">Adam Duracz; Henrik Eriksson; Ferenc A. Bartha; Fei Xu; Yingfu Zeng; Walid Taha</td></tr>
<tr><td>Application Domain</td><td>Automotive</td><td>Aviation/UAV</td><td>Rail</td><td>Medical</td><td>Generic CPS</td><td colspan="3">Other:</td></tr>
<tr><td>Standards Family</td><td>DO</td><td>ISO</td><td>IEC</td><td>EN</td><td colspan="4">Other:</td></tr>
<tr><td>Standards Referenced</td><td colspan="8">ISO 26262</td></tr>
<tr><td>Role of Standard</td><td>Compliance</td><td>Guidance</td><td>Critique</td><td>Mapping</td><td>Terminology only</td><td colspan="3">Other:</td></tr>
<tr><td>Uncertainty Handled</td><td>Explicit</td><td colspan="2">Implicit</td><td>Type of Uncertainty</td><td>Aleatory</td><td>Epistemic</td><td colspan="2">Both</td></tr>
<tr><td>Uncertainty Sources</td><td colspan="8">Sensor, Environment, Model/physics, Software/timing</td></tr>
<tr><td>What is Uncertain?</td><td>Variables</td><td>Parameters</td><td>Assumption</td><td colspan="5">Other:</td></tr>
<tr><td>Representation</td><td>Probabilistic</td><td>Intervals</td><td>Sets</td><td>Scenarios</td><td>Evidence</td><td>Qualitative</td><td colspan="2">Other: Ranking</td></tr>
<tr><td>Quantification method</td><td colspan="3">Reachability under disturbance</td><td>Assumptions</td><td colspan="4">Explicit assumptions; Bounds assumptions</td></tr>
<tr><td>Lifecycle Phases</td><td>Requirements</td><td>Design</td><td>Implementation</td><td>V&V</td><td colspan="2">Runtime/ operations</td><td colspan="2">Other:</td></tr>
<tr><td>Technique Category</td><td colspan="3">Design-time modeling, Assurance/validation</td><td>Technique Name</td><td colspan="4">-</td></tr>
<tr><td>Inputs</td><td colspan="3">Vehicle model</td><td>Outputs</td><td colspan="4">Enclosures</td></tr>
<tr><td>Mitigation Action</td><td colspan="8">Refactor model and tune simulation step-size.</td></tr>
<tr><td>Evidence Maturity Level</td><td colspan="3">Simulation</td><td>Evaluation Setting</td><td colspan="4">Case study</td></tr>
<tr><td>Baselines/ Comparators</td><td>Yes</td><td>No</td><td>Partial</td><td>Dataset / Scenarios</td><td colspan="4">AEBS simulation</td></tr>
<tr><td>Reproducibility</td><td>Yes</td><td>No</td><td>Partial</td><td>Metrics Reported</td><td colspan="4">severity, collision, runtime</td></tr>
</table>

<table>
<tr><td rowspan="2">Applied QARs</td><td>Q1</td><td>Q2</td><td>Q3</td><td>Q4</td><td>Q5</td><td>Q6</td><td>Q7</td><td>Q8</td><td>Q9</td><td>Q10</td></tr>
<tr><td>1</td><td>1</td><td>1</td><td>0.5</td><td>0.5</td><td>0.5</td><td>1</td><td>0.5</td><td>0.5</td><td>0.5</td></tr>
</table>

<table>
<tr><td>Component Focus</td><td>Control</td><td>Safety-Critical scenario</td><td>AEBS</td><td>System Type</td><td>System-level</td></tr>
<tr><td>Contribution of the paper</td><td colspan="5">Applies rigorous simulation to ISO HARA.</td></tr>
<tr><td>Summary</td><td colspan="5">Uses rigorous interval/set-based simulation in Acumen to support ISO 26262 HARA for AEBS, producing bounded collision/severity evidence.</td></tr>
<tr><td>Any Other Comments</td><td colspan="5"></td></tr>
</table>

Study ID	SID4	**Paper Title**	Uncertainty-aware environment simulation of medical devices digital twins							
Publication Year	2024	**Source**	Springer	**Authors**	Hassan Sartaj; Shaukat Ali; Julie Marie Gjøby					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: OMG/OCL					
Standards Referenced	UML, OCL									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Human interaction, Comms/network, Software/timing									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation	**Assumptions**	Explicit assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Architectural techniques	**Technique Name**	EnvDT							
Inputs	UML models	**Outputs**	Simulations							
Mitigation Action	Update digital twin state from events.									
Evidence Maturity Level	Lab-testbed	**Evaluation Setting**	Real hardware							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Dispenser models					
Reproducibility	Yes	No	Partial	**Metrics Reported**	coverage, diversity, simulation time					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0	1	0.5	1	0.5	1	0.5	0.5	0.5
Component Focus	Simulation	**Safety-Critical scenario**	Dispensing	**System Type**	System-level					
Contribution of the paper	Models uncertain medical-device digital-twin environments probabilistically.									
Summary	EnvDT models probabilistic uncertain environments for medical-device digital twins, improving test realism via coverage/diversity analysis on medicine dispensers.									
Any Other Comments										

Study ID	SID5	**Paper Title**	What, Indeed, is an Achievable Provable Guarantee for Learning-Enabled Safety-Critical Systems							
Publication Year	2024	**Source**	Springer	**Authors**	Saddek Bensalem; Chih-Hong Cheng; Wei Huang; Xiaowei Huang; Changshun Wu; Xingyu Zhao					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	IEC 61508, ISO 26262, DO-178C, ISO 21448 (SOTIF)									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Environment, Sensor, ML/data-driven, Model/physics, Software/timing, Security/adversary									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation	**Assumptions**	Explicit assumptions; Distributions assumptions; Independence assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Runtime monitoring, Runtime assurance, Assurance/validation, Formal verification	**Technique Name**	-							
Inputs	Runtime data	**Outputs**	Safety guarantees							
Mitigation Action	Ignore benign errors; switch degraded controller.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Discussion							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Conceptual scenarios					
Reproducibility	Yes	No	Partial	**Metrics Reported**	generalization error, reliability, confidence					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	1	0.5	0.5	0	0	0.5	0.5	0.5
Component Focus	Assurance	**Safety-Critical scenario**	Generic	**System Type**	Hybrid					
Contribution of the paper	Frames statistical guarantees for learning-enabled safety.									
Summary	A methodology paper arguing learning-enabled safety systems need statistical, not absolute, guarantees using ε/δ bounds, runtime monitors, and fallback actions.									
Any Other Comments										

Study ID	SID6	**Paper Title**	Ergo, SMIRK is safe: a safety case for a machine learning component in a pedestrian automatic emergency brake system							
Publication Year	2023	**Source**	Springer	**Authors**	Markus Borg; Jens Henriksson; Kasper Socha; Olof Lennartsson; Elias Sonnsjö Lönegren; Thanh Bui; Piotr Tomaszewski; Sankar Raman Sathyamoorthy; Sebastian Brink; Mahshid Helali Moghadam					
Application Domain	==Automotive==	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	==ISO==	IEC	EN	==Other: IEEE, SAE==					
Standards Referenced	ISO 21448 (SOTIF), ISO 26262 , SAE J3016 , IEEE 830-1998									
Role of Standard	Compliance	==Guidance==	Critique	==Mapping==	Terminology only	Other:				
Uncertainty Handled	==Explicit==	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	==Both==				
Uncertainty Sources	Environment, Sensor, ML/data-driven, Human interaction, Comms/network									
What is Uncertain?	==Variables==	Parameters	==Assumption==	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	==Other: Ranking==			
Quantification method	Calibration; Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	==Requirements==	==Design==	==Implementation==	==V&V==	==Runtime/ operations==	Other:				
Technique Category	Assurance/validation, Runtime monitoring, Runtime assurance, Architectural techniques	**Technique Name**	SMIRK							
Inputs	SMIRK data	**Outputs**	Safety evidence							
Mitigation Action	Block braking and hand control to driver.									
Evidence Maturity Level	Simulation , Industrial case	**Evaluation Setting**	Case study , Simulator/tool							
Baselines/ Comparators	Yes	==No==	Partial	**Dataset / Scenarios**	SMIRK dataset					
Reproducibility	==Yes==	No	Partial	**Metrics Reported**	precision, recall, F1, AP, FPPI, errors, collisions, inference time					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	1	1	1	1	0.5	1	0.5
Component Focus	Perception	**Safety-Critical scenario**	PAEB	**System Type**	System-level					
Contribution of the paper	Publishes AMLAS safety case for SMIRK.									
Summary	Builds a full AMLAS/SOTIF-style safety case for SMIRK PAEB, using YOLOv5 plus autoencoder OOD detection to avoid ghost braking.									
Any Other Comments										

Study ID	SID7	**Paper Title**	VISEF: Empirical Evaluation of a Vehicle-Level Integrated Functional Safety Framework							
Publication Year	2025	**Source**	Springer	**Authors**	Taewan Gu					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: SAE					
Standards Referenced	SAE J1939, ISO 26262									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Model/physics, Software/timing, Sensor, Environment, Human interaction									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	UQ propagation; Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other: Feedback				
Technique Category	Design-time modeling, Assurance/validation, Architectural techniques	**Technique Name**	VISEF							
Inputs	Hazard data	**Outputs**	Hazard traceability							
Mitigation Action	Reassign ASILs, add diagnostics, rerun HARA.									
Evidence Maturity Level	Simulation , Industrial case	**Evaluation Setting**	Simulator/tool, Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Monte Carlo					
Reproducibility	Yes	No	Partial	**Metrics Reported**	hazard detection, stats tests, entropy, inconsistencies					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	1	0.5	1	0.5	1	1	0.5
Component Focus	Assurance	**Safety-Critical scenario**	Cross-domain	**System Type**	System-level					
Contribution of the paper	Introduces vehicle-level ISO 26262 integration framework.									
Summary	VISEF extends ISO 26262 from item-level to vehicle-level safety by integrating hazards and checking ASIL consistency across ECU safety paths.									
Any Other Comments										

Study ID	SID8	**Paper Title**	The black-box simplex architecture for runtime assurance of multi-agent CPS							
Publication Year	2024	**Source**	Springer	**Authors**	Sanaz Sheikhi; Usama Mehmood; Stanley Bak; Scott A. Smolka; Scott D. Stoller					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	RTCA (DO-386)									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Environment									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Reachability under disturbance	**Assumptions**	Explicit assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Runtime assurance, Architectural techniques	**Technique Name**	Black-Box Simplex Architecture (BSA)							
Inputs	System model	**Outputs**	Safe commands							
Mitigation Action	Switch to stored safe command sequence.									
Evidence Maturity Level	Simulation , Lab-testbed	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Collision avoidance					
Reproducibility	Yes	No	Partial	**Metrics Reported**	safety violations, separation, transparency, runtime					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	0.5	1	1	1	1	1	1	0.5
Component Focus	Assurance	**Safety-Critical scenario**	Collision	**System Type**	System-level					
Contribution of the paper	Removes verified-baseline requirement from Simplex assurance.									
Summary	Black-Box Simplex provides runtime assurance for autonomous CPS by checking controller commands online and falling back to stored safe command sequences.									
Any Other Comments										

Study ID	SID9	**Paper Title**	Correct-by-construction requirement decomposition							
Publication Year	2025	**Source**	Springer	**Authors**	Minghui Sun; Georgios Bakirtzis; Hassan Jafarzadeh; Cody Fleming					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: SAE, IEEE/ISO/IEC					
Standards Referenced	SAE (EIA-632), ARP4754B, IEEE/ISO/IEC 29148:2018									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Model/physics, Environment									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Reachability under disturbance; Worst-case bounds	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Formal verification, Architectural techniques, Assurance/validation	**Technique Name**	Correct-by-construction requirement decomposition (CBD)							
Inputs	Contracts	**Outputs**	Bounded contracts							
Mitigation Action	Revise design; resolve supplier capability conflicts.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Cruise control					
Reproducibility	Yes	No	Partial	**Metrics Reported**	set satisfaction, feasibility, bounds					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	1	1	0.5	1	0.5	0.5	0.5
Component Focus	Requirements	**Safety-Critical scenario**	Cruise-control	**System Type**	System-level					
Contribution of the paper	Automatically derives correct-by-construction subrequirements using contracts.									
Summary	Decomposes high-level requirements into correct-by-construction sub-requirements using contracts, reachability, CSP, and optimization under uncertainty.									
Any Other Comments										

Study ID	SID10	**Paper Title**	U-Map: A Reference Map for Safe Handling of Runtime Uncertainties							
Publication Year	2020	**Source**	Springer	**Authors**	Nishanth Laxman; Chee Hung Koo; Peter Liggesmeyer					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other: Industrial				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 12100:2010									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Human interaction, Comms/network, Sensor									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	UQ propagation	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Runtime monitoring, Runtime assurance	**Technique Name**	U-Map							
Inputs	Uncertainty evidence	**Outputs**	Mitigation map							
Mitigation Action	Rectify hazards, degrade speed, raise alarms.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Industrial integration					
Reproducibility	Yes	No	Partial	**Metrics Reported**	hazard probability, accident likelihood					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Component Focus	Assurance	**Safety-Critical scenario**	Robotics	**System Type**	System-level					
Contribution of the paper	Maps runtime uncertainties to hazard mitigations.									
Summary	U-Map maps runtime uncertainties to hazards and mitigations, using monitoring and Bayesian reasoning to reduce accident likelihood in industrial CPS.									
Any Other Comments										

Study ID	SID11	**Paper Title**	Formally verifying decompositions of stochastic specifications							
Publication Year	2024	**Source**	Springer	**Authors**	Anton Hampus; Mattias Nyberg					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 26262, ISO 21434									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Model/physics, Software/timing									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Worst-case bounds		**Assumptions**	Explicit assumptions; Bounds assumptions						
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Formal verification, Assurance/validation		**Technique Name**	Probabilistic Automaton Contracts (PACs)						
Inputs	PAC specs		**Outputs**	Refinement verdicts						
Mitigation Action	Verify specifications; no runtime mitigation provided.									
Evidence Maturity Level	Conceptual		**Evaluation Setting**	Case study						
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Power backup					
Reproducibility	Yes	No	Partial	**Metrics Reported**	soundness, completeness, path size					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	1	0.5	0.5	0.5	0.5	0.5
Component Focus	Assurance	**Safety-Critical scenario**	Power	**System Type**	System-level					
Contribution of the paper	Formalizes stochastic contract decomposition verification semantics.									
Summary	Formalizes stochastic requirement decomposition using probabilistic automaton contracts and refinement checks via linear constraints over trace probabilities.									
Any Other Comments										

Study ID	SID12	**Paper Title**	A Modular SystemC RTOS Model for Uncertainty Analysis							
Publication Year	2020	**Source**	Springer	**Authors**	Lorenzo Lazzara; Giulio Mosè Mancuso; Fabio Cremona; Alessandro Ulisse					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: ARINC, ISO/IEC, IEEE, AUTOSAR					
Standards Referenced	ARINC 653, ISO/IEC 14882:2017, POSIX (IEEE 1003.1), OSEK OS (ISO 17356)									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Software/timing, Environment									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	UQ propagation; Statistical estimation		**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions						
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Assurance/validation, Formal verification		**Technique Name**	-						
Inputs	RTOS config		**Outputs**	Timing probabilities						
Mitigation Action	Assess timing; no runtime mitigation proposed.									
Evidence Maturity Level	Simulation		**Evaluation Setting**	Simulator/tool						
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Task sets					
Reproducibility	Yes	No	Partial	**Metrics Reported**	response time, deadline miss, BLTL satisfaction, simulation cost					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	1	0.5	1	0.5	1	0.5	0.5	0.5
Component Focus	Scheduling	**Safety-Critical scenario**	Avionics	**System Type**	Hybrid					
Contribution of the paper	Builds configurable SystemC RTOS uncertainty model.									
Summary	Presents a SystemC RTOS simulation model for uncertainty analysis, showing small timing uncertainties can cause major response-time violations.									
Any Other Comments										

Study ID	SID13	**Paper Title**	Autonomous Vehicles Meet the Physical World: RSS, Variability, Uncertainty, and Proving Safety							
Publication Year	2019	**Source**	Springer	**Authors**	Philip Koopman; Beth Osyk; Jack Weast					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: SAE					
Standards Referenced	ISO/PAS 21448, SAE J3016									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, Human interaction									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Worst-case bounds; UQ propagation	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Formal verification, Design-time modeling, Runtime assurance, Architectural techniques, Assurance/validation	**Technique Name**	Micro-Operational Design Domain (μODD)							
Inputs	Vehicle kinematics	**Outputs**	Safety envelope							
Mitigation Action	Transition to conservative micro-ODD or stop.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	AV scenarios					
Reproducibility	Yes	No	Partial	**Metrics Reported**	safety distance, permissiveness					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	1	0.5	0.5	0.5	0.5	1	0.5
Component Focus	Assurance	**Safety-Critical scenario**	Braking	**System Type**	System-level					
Contribution of the paper	Extends RSS for uncertainty-bounded ODD slices.									
Summary	Shows RSS-style AV safety becomes brittle under real-world variability, then proposes bounded μODD slices and conservative fallback when confidence drops.									
Any Other Comments										

Study ID	SID14	**Paper Title**	Advancing real-time validation of automotive software systems via continuous integration and intelligent failure analysis							
Publication Year	2025	**Source**	Springer	**Authors**	Mohammad Abboush; Christoph Knieke; Andreas Rausch					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 26262									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, Software/timing, ML/data-driven, Comms/network									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation; Calibration	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Architectural techniques, Design-time modeling	**Technique Name**	GRU-DAE							
Inputs	HIL records	**Outputs**	Fault reports							
Mitigation Action	Notify developers through traceable validation reports.									
Evidence Maturity Level	Lab-testbed	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Fault injection					
Reproducibility	Yes	No	Partial	**Metrics Reported**	precision, recall, F1, MSE, DBI, CI duration					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	0.5	1	1	1	1	0.5	0.5
Component Focus	Validation	**Safety-Critical scenario**	Powertrain	**System Type**	System-level					
Contribution of the paper	Automates HIL validation with ML diagnostics.									
Summary	Turns automotive HIL validation into a CI pipeline and uses ML-based failure diagnosis to improve noisy fault classification and reduce test turnaround.									
Any Other Comments										

Study ID	SID15	**Paper Title**	Real-time multi-agent systems: rationality, formal model, and empirical results							
Publication Year	2021	**Source**	Springer	**Authors**	Davide Calvaresi; Yashin Dicente Cid; Mauro Marinoni; Aldo Franco Dragoni; Amro Najjar; Michael Schumacher					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: IEEE, OMG, JCP					
Standards Referenced	IEEE (FIPA), OMG (CORBA) also ISO/IEC 19500, OMG (Real-Time CORBA)									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Software/timing, Comms/network									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Formal			
Quantification method	Worst-case bounds; Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Assurance/validation, Architectural techniques	**Technique Name**	Reservation Based Negotiation (RBN)							
Inputs	Task-sets	**Outputs**	Timing schedules							
Mitigation Action	Reject overload; postpone deadlines after overruns.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Task sets					
Reproducibility	Yes	No	Partial	**Metrics Reported**	deadline miss					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	0.5	1	1	1	1	1	0.5	0.5
Component Focus	Scheduling	**Safety-Critical scenario**	Generic	**System Type**	Component-level					
Contribution of the paper	Formally defines schedulability-aware real-time multi-agent systems.									
Summary	Formalizes real-time multi-agent systems with schedulability-aware negotiation, showing deadline misses drop to zero in tested RT-MAS simulations.									
Any Other Comments										

Study ID	SID16	**Paper Title**	RTOS Solution for NoC-Based COTS MPSoC Usage in Mixed-Criticality Systems				
Publication Year	2019	**Source**	Springer	**Authors**	Serhiy Avramenko; Massimo Violante		
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:	
Standards Family	DO	ISO	IEC	EN	Other: ARINC, SAE, IEEE		
Standards Referenced	DO-178C, DO-254, ISO 26262, IEC 61508, ARINC 653, SAE ARP4761, IEEE Std 1228						
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:	
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both	
Uncertainty Sources	Comms/network, Software/timing, Security/adversary						
What is Uncertain?	Variables	Parameters	Assumption	Other:			
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical
Quantification method	Statistical estimation; Worst-case bounds		**Assumptions**	Explicit assumptions			
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:	
Technique Category	Architectural techniques, Runtime monitoring, Runtime assurance, Assurance/validation		**Technique Name**	-			
Inputs	NoC topology		**Outputs**	Traffic filtering			
Mitigation Action	Drop violating packets; reroute noncritical traffic.						
Evidence Maturity Level	Simulation		**Evaluation Setting**	Simulator/tool			
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	NoC traffic		
Reproducibility	Yes	No	Partial	**Metrics Reported**	latency, reachability, connectivity		

Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	1	1	0.5	1	1	0.5	0.5

Component Focus	RTOS	**Safety-Critical scenario**	Avionics	**System Type**	Component-level
Contribution of the paper	Enforces NoC isolation through RTOS filtering.				
Summary	Adds RTOS-level NoC traffic filtering for mixed-criticality MPSoCs to prevent inter-criticality interference and make critical latency deterministic.				
Any Other Comments					

Study ID	SID17	**Paper Title**	Towards Integrated Quantitative Security and Safety Risk Assessment							
Publication Year	2019	**Source**	Springer	**Authors**	Jürgen Dobaj; Christoph Schmittner; Michael Krisper; Georg Macher					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 31000, ISO 26262, ISO/IEC 15408, IEC 60812, ISO 27005									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Security/adversary, Human interaction, Software/timing, Comms/network									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation; UQ propagation		**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions; Independence assumptions						
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Assurance/validation		**Technique Name**	-						
Inputs	Threat models		**Outputs**	Security risk						
Mitigation Action	Add mitigation models for risk treatment.									
Evidence Maturity Level	Conceptual		**Evaluation Setting**	Illustrative Example						
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	-					
Reproducibility	Yes	No	Partial	**Metrics Reported**	risk probability					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	0.5	0	0	0.5	0.5	0.5
Component Focus	Risk	**Safety-Critical scenario**	Steering	**System Type**	System-level					
Contribution of the paper	Integrates contracts and self-management for COTS.									
Summary	Combines Diamond attack modeling and FAIR-style probabilistic risk estimation for more quantitative integrated safety-security risk assessment.									
Any Other Comments										

<table>
<tr><td>Study ID</td><td>SID18</td><td>Paper Title</td><td colspan="5">A Methodological Framework for Model-Based Self-management of Services and Components in Dependable Cyber-Physical Systems</td></tr>
<tr><td>Publication Year</td><td>2018</td><td>Source</td><td>Springer</td><td>Authors</td><td colspan="3">DeJiu Chen; Zhonghai Lu</td></tr>
<tr><td>Application Domain</td><td>Automotive</td><td>Aviation/UAV</td><td>Rail</td><td>Medical</td><td>Generic CPS</td><td colspan="2">Other:</td></tr>
<tr><td>Standards Family</td><td>DO</td><td>ISO</td><td>IEC</td><td>EN</td><td colspan="3">Other: SAE, OMG, SEI, AUTOSAR</td></tr>
<tr><td>Standards Referenced</td><td colspan="7">ISO 26262, , SAE J3016, OMG SysML also ISO/IEC 19514:2017, SEI AADL (SAE (AS5506))</td></tr>
<tr><td>Role of Standard</td><td>Compliance</td><td>Guidance</td><td>Critique</td><td>Mapping</td><td>Terminology only</td><td colspan="2">Other:</td></tr>
<tr><td>Uncertainty Handled</td><td>Explicit</td><td>Implicit</td><td>Type of Uncertainty</td><td>Aleatory</td><td>Epistemic</td><td colspan="2">Both</td></tr>
<tr><td>Uncertainty Sources</td><td colspan="7">Sensor, Environment, Software/timing, Comms/network</td></tr>
<tr><td>What is Uncertain?</td><td>Variables</td><td>Parameters</td><td>Assumption</td><td colspan="4">Other:</td></tr>
<tr><td>Representation</td><td>Probabilistic</td><td>Intervals</td><td>Sets</td><td>Scenarios</td><td>Evidence</td><td>Qualitative</td><td>Other: Formal</td></tr>
<tr><td>Quantification method</td><td colspan="3">Probabilistic adaptation</td><td>Assumptions</td><td colspan="3">Explicit assumptions; Bounds assumptions</td></tr>
<tr><td>Lifecycle Phases</td><td>Requirements</td><td>Design</td><td>Implementation</td><td>V&V</td><td>Runtime/ operations</td><td colspan="2">Other:</td></tr>
<tr><td>Technique Category</td><td colspan="3">Design-time modeling, Runtime monitoring, Runtime assurance</td><td>Technique Name</td><td colspan="3">-</td></tr>
<tr><td>Inputs</td><td colspan="3">EAST-ADL</td><td>Outputs</td><td colspan="3">Adaptation contracts</td></tr>
<tr><td>Mitigation Action</td><td colspan="7">Trigger error handling and configuration adaptation.</td></tr>
<tr><td>Evidence Maturity Level</td><td colspan="3">Conceptual</td><td>Evaluation Setting</td><td colspan="3">Simulator/tool</td></tr>
<tr><td>Baselines/ Comparators</td><td>Yes</td><td>No</td><td>Partial</td><td>Dataset / Scenarios</td><td colspan="3">Braking system</td></tr>
<tr><td>Reproducibility</td><td>Yes</td><td>No</td><td>Partial</td><td>Metrics Reported</td><td colspan="3">-</td></tr>
</table>

Applied QARs	Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
	1	0.5	0.5	0.5	0.5	0	0.5	0	0.5	0.5

<table>
<tr><td>Component Focus</td><td>Assurance</td><td>Safety-Critical scenario</td><td>Braking</td><td>System Type</td><td>System-level</td></tr>
<tr><td>Contribution of the paper</td><td colspan="5">Integrates contracts and self-management for COTS.</td></tr>
<tr><td>Summary</td><td colspan="5">Uses EAST-ADL and assume–guarantee contracts to monitor CPS service/component dependability and adapt configurations at runtime.</td></tr>
<tr><td>Any Other Comments</td><td colspan="5"></td></tr>
</table>

Study ID	SID19	**Paper Title**	A Model-Based Approach to Dynamic Self-assessment for Automated Performance and Safety Awareness of Cyber-Physical Systems							
Publication Year	2017	**Source**	Springer	**Authors**	DeJiu Chen; Zhonghai Lu					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: SAE					
Standards Referenced	ISO 26262, SAE J3016									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Software/timing, Comms/network, ML/data-driven									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Formal			
Quantification method	Statistical estimation; UQ propagation	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions; Independence assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other: Maintenance				
Technique Category	Design-time modeling, Runtime monitoring, Runtime assurance, Architectural techniques, Assurance/validation	**Technique Name**	MAS/SAS services							
Inputs	Contracts	**Outputs**	State estimates							
Mitigation Action	Adapt configuration; switch modes, treat faults.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Illustrative Example, Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Observation sequences					
Reproducibility	Yes	No	Partial	**Metrics Reported**	state probability, violation risk					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	0.5	0	0.5	0.5	1	0.5
Component Focus	Monitoring	**Safety-Critical scenario**	Braking	**System Type**	System-level					
Contribution of the paper	Adds HMM self-assessment for operational uncertainty.									
Summary	Extends model-based CPS self-assessment with runtime contracts and HMM-based hidden-state inference to drive QoS adaptation and risk mitigation.									
Any Other Comments										

Study ID	SID20	**Paper Title**	Trustworthiness modeling and evaluation for a nearly autonomous management and control system							
Publication Year	2024	**Source**	Science Direct	**Authors**	Longcong Wang; Linyu Lin; Nam Dinh					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other: Nuclear				
Standards Family	DO	ISO	IEC	EN	Other: SCSC					
Standards Referenced	SCSC-141C									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Model/physics, ML/data-driven, Human interaction, Software/timing									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	UQ propagation; Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions; Independence assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other: Maintenance				
Technique Category	Design-time modeling, Runtime monitoring, Runtime assurance, Assurance/validation	**Technique Name**	-							
Inputs	Trust evidence	**Outputs**	Trustworthiness scores							
Mitigation Action	SCRAM reactor when discrepancy exceeds limits.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	NAMAC scenarios					
Reproducibility	Yes	No	Partial	**Metrics Reported**	correctness confidence, PPV, satisfaction, sensitivity					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	0.5	1	1	0.5	0.5	0.5	0.5	0.5
Component Focus	DigitalTwin	**Safety-Critical scenario**	Reactor	**System Type**	System-level					
Contribution of the paper	Models NAMAC trustworthiness using Bayesian ontologies.									
Summary	Models trustworthiness for nuclear control CPS using ontologies, Bayesian networks, evidence, and sensitivity analysis over uncertain assumptions.									
Any Other Comments										

Study ID	SID21	**Paper Title**	Non-deterministic model validation methodology for simulation-based safety assessment of automated vehicles						
Publication Year	2021	**Source**	Science Direct	**Authors**	Stefan Riedmaier; Jakob Schneider; Benedikt Danquah; Bernhard Schick; Frank Diermeyer				
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:			
Standards Family	DO	ISO	IEC	EN	Other: SAE, ANSI-UL, ASME				
Standards Referenced	SAE J3016, UL 4600, ISO 19365:2016, ISO 19364:2016, ASME V\&V 20-2009								
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:			
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both			
Uncertainty Sources	Sensor, Model/physics, Environment, Software/timing								
What is Uncertain?	Variables	Parameters	Assumption	Other:					
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical		
Quantification method	UQ propagation; Statistical estimation		**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions					
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:			
Technique Category	Design-time modeling, Assurance/validation		**Technique Name**	-					
Inputs	AV model		**Outputs**	Type approval					
Mitigation Action	Adjust certification decision using inferred uncertainties.								
Evidence Maturity Level	Simulation		**Evaluation Setting**	Simulator/tool					
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	DoE scenarios				
Reproducibility	Yes	No	Partial	**Metrics Reported**	line distance, deviation, validation, precision, recall				

Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	1	1	1	1	1	1	0.5	0.5

Component Focus	Control	**Safety-Critical scenario**	Lanekeeping	**System Type**	System-level
Contribution of the paper	Validates AV simulations with non-deterministic UQ.				
Summary	Proposes non-deterministic simulation validation for AV safety assessment using p-boxes and model-error expansion to reduce false safety conclusions.				
Any Other Comments					

Study ID	SID22	**Paper Title**	A Probabilistic Approach to Adaptive Protection in the Smart Grid							
Publication Year	2025	**Source**	ACM	**Authors**	Amr S. Mohamed; Deepa Kundur; Mohsen Khalaf					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other: Smart Grid				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	IEC 60909, IEC 61850									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Environment, Model/physics, Comms/network									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	UQ propagation; Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Assurance/validation, Architectural techniques	**Technique Name**	-							
Inputs	Grid model	**Outputs**	Relay settings							
Mitigation Action	Switch relay mode; trip backup breakers.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Simulator/tool, Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Fault datasets					
Reproducibility	Yes	No	Partial	**Metrics Reported**	accuracy, operating time					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	1	1	1	0.5	0.5	0.5
Component Focus	Protection	**Safety-Critical scenario**	Short-circuit	**System Type**	System-level					
Contribution of the paper	Applies interpretable probabilistic adaptive grid protection.									
Summary	Uses probabilistic ML and CVaR optimization for adaptive smart-grid protection under uncertain fault conditions, validated on CIGRE MV cases.									
Any Other Comments										

Study ID	SID23	**Paper Title**	A Reference Architecture of Human Cyber-Physical Systems – Part III: Semantic Foundations							
Publication Year	2024	**Source**	ACM	**Authors**	Werner Damm; Martin Fränzle; Alyssa J. Kerscher; Forrest Laine; Klaus Bengler; Bianca Biebl; Willem Hagemann; Moritz Held; David Hess; Klas Ihme; Severin Kacianka; Sebastian Lehnhoff; Andreas Luedtke; Alexander Pretschner; Astrid Rakow; Jochem Rieger; Daniel Sonntag; Janos Sztipanovits; Maike Schwammberger; Mark Schweda; Alexander Trende; Anirudh Unni; Eric Veith					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: SAE					
Standards Referenced	SAE ARP4754A									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, Software/timing, Human interaction, ML/data-driven, Comms/network, Security/adversary									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation; UQ propagation	**Assumptions**	Explicit assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Formal verification, Assurance/validation, Architectural techniques	**Technique Name**	RA(HCPS)							
Inputs	Agent models	**Outputs**	Semantic artifacts							
Mitigation Action	Update strategy and coalition from beliefs.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Case study , Illustrative Example							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Cockpit examples					
Reproducibility	Yes	No	Partial	**Metrics Reported**	risk levels					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Component Focus	Assurance	**Safety-Critical scenario**	Aviation	**System Type**	System-level					
Contribution of the paper	Formalizes semantic foundations for human-CPS beliefs.									
Summary	Provides probabilistic game-theoretic semantics for human CPS, modeling belief inconsistency and uncertainty in aviation/automotive/UAV contexts.									
Any Other Comments										

Study ID	SID24	**Paper Title**	Investigating the Inherent Soft Error Resilience of Embedded Applications by Full-System Simulation							
Publication Year	2020	**Source**	ACM	**Authors**	Uzair Sharif; Daniel Mueller-Gritschneder; Ulf Schlichtmann					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other: Industrial				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 26262									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, Software/timing									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical			
Quantification method	Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Design-time modeling	**Technique Name**	-							
Inputs	Virtual prototypes	**Outputs**	Fault classifications							
Mitigation Action	Exploit inherent resilience; no runtime response.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	ACC robot					
Reproducibility	Yes	No	Partial	**Metrics Reported**	safety violations, behavior deviation, severity					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	0.5	0.5	0.5	0.5	0.5	0.5
Component Focus	Control	**Safety-Critical scenario**	ACC	**System Type**	System-level					
Contribution of the paper	Quantifies embedded resilience via full-system simulation.									
Summary	Uses full-system simulation to classify soft-error impacts by actual closed-loop safety behavior, avoiding over-protection of harmless SDCs.									
Any Other Comments										

Study ID	SID25	**Paper Title**	Oracles for Testing Software Timeliness with Uncertainty							
Publication Year	2018	**Source**	ACM	**Authors**	Chunhui Wang; Fabrizio Pastore; Lionel Briand					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 26262									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Environment, Sensor, Software/timing, Comms/network									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions; Independence assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Formal verification, Design-time modeling	**Technique Name**	STUIOS, PUIO							
Inputs	Timed automata	**Outputs**	Test verdicts							
Mitigation Action	Fail oracle or flag unreliable verdicts.									
Evidence Maturity Level	Industrial case	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Mutated specs					
Reproducibility	Yes	No	Partial	**Metrics Reported**	precision, recall, fault detection, execution cost, oracle accuracy					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	1	1	1	1	1	1	0.5
Component Focus	Oracles	**Safety-Critical scenario**	Airbag	**System Type**	Component-level					
Contribution of the paper	Reliably generates stochastic timing-test oracles automatically.									
Summary	Builds probabilistic timing-test oracles using UPPAAL SMC and repeated-run hypothesis testing to handle nondeterministic embedded timing behavior.									
Any Other Comments										

Study ID	SID26	**Paper Title**	Dependable Model-driven Development of CPS: From Stateflow Simulation to Verified Implementation							
Publication Year	2018	**Source**	ACM	**Authors**	Yu Jiang; Houbing Song; Yixiao Yang; Han Liu; Ming Gu; Yong Guan; Jiaguang Sun; Lui Sha					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	IEC 61375, IEC-61375-1, , IEC-61375-2									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Comms/network, Software/timing, Environment									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Formal			
Quantification method	Reachability under disturbance; Worst-case bounds	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Formal verification, Runtime monitoring, Assurance/validation	**Technique Name**	STU, RMOR, DRTV							
Inputs	Simulink models	**Outputs**	Runtime monitors							
Mitigation Action	Revise design, add handshake, harden notifications.									
Evidence Maturity Level	Deployed	**Evaluation Setting**	Simulator/tool, Real hardware							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Injected faults					
Reproducibility	Yes	No	Partial	**Metrics Reported**	defects, false positives, verification time, delay					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	0.5	1	1	0.5	1	1	0.5
Component Focus	Communication	**Safety-Critical scenario**	Train	**System Type**	System-level					
Contribution of the paper	Links Stateflow models to verification monitors.									
Summary	Links Stateflow simulation to verified implementation using Uppaal model checking and runtime monitoring, catching defects and timing violations on train hardware.									
Any Other Comments										

Study ID	SID27	**Paper Title**	Automated approach for system-level testing of unmanned aerial systems							
Publication Year	2022	**Source**	ACM	**Authors**	Hassan Sartaj					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: STANAG					
Standards Referenced	DO-178C, STANAG 4586									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Environment, Model/physics, Software/timing, Comms/network									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Violation-count based reward		**Assumptions**	Explicit assumptions						
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Design-time modeling, Runtime monitoring		**Technique Name**	AITester						
Inputs	UAV models		**Outputs**	Test scenarios						
Mitigation Action	Report constraint violations; no mitigation implemented.									
Evidence Maturity Level	Simulation		**Evaluation Setting**	Simulator/tool						
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Online scenarios					
Reproducibility	Yes	No	Partial	**Metrics Reported**	OCL failures, accuracy, execution time					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	0.5	0.5	0	0.5	0.5	0.5	1	1	0.5	0.5
Component Focus	Testing	**Safety-Critical scenario**	Autopilot	**System Type**	System-level					
Contribution of the paper	Combines MBT and deep-RL UAS testing.									
Summary	AITester uses UML/OCL specifications plus DRL to generate UAS simulator scenarios that maximize runtime constraint violations.									
Any Other Comments										

Study ID	SID28	**Paper Title**	A Hybrid Target Selection Model of Functional Safety Compliance for Autonomous Driving System				
Publication Year	2025	**Source**	ACM	**Authors**	Yang Liu; Zihan Wang; Mengchi Cai; Qing Xu; Keqiang Li		
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:	
Standards Family	DO	ISO	IEC	EN	Other:		
Standards Referenced	ISO 26262						
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:	
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both	
Uncertainty Sources	Sensor, Environment, ML/data-driven, Human interaction, Software/timing						
What is Uncertain?	Variables	Parameters	Assumption	Other:			
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical
Quantification method	Statistical estimation; Calibration		**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions			
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:	
Technique Category	Architectural techniques, Runtime monitoring, Design-time modeling		**Technique Name**	-			
Inputs	Object features		**Outputs**	Target selection			
Mitigation Action	Debounce disagreement, then apply thresholded fallback.						
Evidence Maturity Level	Industrial case		**Evaluation Setting**	Simulator/tool, Real hardware			
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Road data		
Reproducibility	Yes	No	Partial	**Metrics Reported**	precision, accuracy, recall, specificity, F1, AUC, confusion counts, runtime		

Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	0.5	0.5	0.5	0.5	0.5	1	1	0.5

Component Focus	Perception	**Safety-Critical scenario**	Target-selection	**System Type**	Component-level
Contribution of the paper	Hybridizes LSTM with safety-certified expert rules.				
Summary	Combines LSTM target selection with rule-based expert logic and probabilistic smoothing for functionally safer autonomous-driving target selection.				
Any Other Comments					

Study ID	SID29	**Paper Title**	Early Reliability Assessment of AI-based Automotive Systems							
Publication Year	2025	**Source**	ACM	**Authors**	Shailesh Hegde; Dinesh Selvaraj; Josie Esteban Rodriguez Condia; Nicola Amati; C.-F. Chiasserini; Francesco Deflorio; Matteo Sonza Reorda					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: ISO/IEC					
Standards Referenced	ISO 26262, ISO/IEC 22989, ISO 2631-1:1997, BS ISO 8855:2011, BS ISO 34502:2022									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, Software/timing, ML/data-driven									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation; Worst-case bounds		**Assumptions**	Explicit assumptions; Bounds assumptions						
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Design-time modeling		**Technique Name**	TIARA						
Inputs	Fault scenarios		**Outputs**	Fault rankings						
Mitigation Action	Harden sensitive regions; validate critical scenarios.									
Evidence Maturity Level	Simulation , Lab-testbed		**Evaluation Setting**	Simulator/tool, Real hardware						
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Camera simulation					
Reproducibility	Yes	No	Partial	**Metrics Reported**	lane changes, acceleration, steering, deviation, comfort, execution time					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	0.5	0.5	1	1	1	1	0.5	0.5
Component Focus	Perception	**Safety-Critical scenario**	Lanekeeping	**System Type**	Hybrid					
Contribution of the paper	Introduces TIARA for AI automotive reliability.									
Summary	TIARA performs early reliability assessment of AI automotive systems by narrowing NN fault analysis to critical faults, then validating in simulation/HIL.									
Any Other Comments										

Study ID	SID30	**Paper Title**	Analyzing execution path non-determinism of the Linux kernel in different scenarios							
Publication Year	2023	**Source**	Taylor and Francis	**Authors**	Yucong Chen; Xianzhi Tang; Shuaixin Xu; Fangfang Zhu; Qingguo Zhou; Tien-Hsiung Weng					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	IEC 61508									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Software/timing									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical			
Quantification method	Statistical estimation	**Assumptions**	Explicit assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation	**Technique Name**	-							
Inputs	Kernel traces	**Outputs**	Path hashes							
Mitigation Action	Analyze paths; plan verification, not runtime.									
Evidence Maturity Level	Lab-testbed	**Evaluation Setting**	Real hardware							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Execution paths					
Reproducibility	Yes	No	Partial	**Metrics Reported**	path counts, path frequency, switches, rare paths					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	1	1	1	1	0.5	0.5
Component Focus	OS	**Safety-Critical scenario**	Certification	**System Type**	Component-level					
Contribution of the paper	Empirically traces Linux execution-path non-determinism online.									
Summary	Measures Linux kernel execution-path nondeterminism under different loads/filesystems, showing rare path variability complicates certification coverage.									
Any Other Comments										

Study ID	SID31	**Paper Title**	Execution time analysis and optimisation techniques in the model-based development of a flight control software							
Publication Year	2017	**Source**	Wiley	**Authors**	Kajetan Nürnberger; Markus Hochstrasser; Florian Holzapfel					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: STANAG					
Standards Referenced	DO-178C, DO-331, DO-248C, DO-254 , STANAG 4671									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Software/timing, Model/physics, Environment									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Worst-case bounds		**Assumptions**	Explicit assumptions; Bounds assumptions						
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Architectural techniques, Optimization, Formal verification		**Technique Name**	-						
Inputs	Flight model		**Outputs**	WCET bounds						
Mitigation Action	Restructure models to reduce WCET overestimation.									
Evidence Maturity Level	Lab-testbed		**Evaluation Setting**	Case study						
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Operational modes					
Reproducibility	Yes	No	Partial	**Metrics Reported**	execution time, WCET, overestimation					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	1	1	1	1	1	1	0.5
Component Focus	Control	**Safety-Critical scenario**	Flight-control	**System Type**	System-level					
Contribution of the paper	Optimizes flight-control WCET through modeling choices.									
Summary	Compares flight-control build/compiler variants using HIL timing measurement and static WCET analysis, recommending model/code structures to reduce overestimation.									
Any Other Comments										

Study ID	SID32	**Paper Title**	Assessing the Usefulness of Assurance Cases: Experience With the Large Hadron Collider							
Publication Year	2025	**Source**	Wiley	**Authors**	Torin Viger; Jeff Joyce; Simon Diemert; Claudio Menghi; Marsha Chechik; Jan Uythoven; Markus Zerlauth; Lukas Felsberger					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other: Nuclear				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 15026-2, ISO 26262, EN 50126									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, Software/timing, Comms/network									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Empirical metrics	**Assumptions**	Explicit assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation	**Technique Name**	Eliminative Argumentation (EA)							
Inputs	CERN docs	**Outputs**	Assurance case							
Mitigation Action	Trigger beam dump; expand assurance evidence.									
Evidence Maturity Level	Industrial case	**Evaluation Setting**	Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Defeater set					
Reproducibility	Yes	No	Partial	**Metrics Reported**	node counts, defeaters, residual risks, workdays, changes, precision, recall					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	1	1	1	0.5	1	1
Component Focus	Assurance	**Safety-Critical scenario**	Beam-dump	**System Type**	System-level					
Contribution of the paper	Evaluates assurance-case usefulness using LHC evidence.									
Summary	Evaluates assurance-case usefulness on CERN's LHC protection system, showing eliminative argumentation exposed real risk defeaters with high expert-confirmed recall.									
Any Other Comments										

Study ID	SID33	**Paper Title**	Configuration Testing of an Artificial Pancreas System Using a Digital Twin: An Evaluative Case Study							
Publication Year	2025	**Source**	Wiley	**Authors**	Richard Somers; Neil Walkinshaw; Robert Mark Hierons; Jackie Elliott; Ahmed Iqbal; Emma Walkinshaw					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 14971									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Human interaction, Model/physics, ML/data-driven, Software/timing									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical			
Quantification method	Calibration; Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Design-time modeling, Assurance/validation	**Technique Name**	-							
Inputs	OpenAPS traces	**Outputs**	Simulation results							
Mitigation Action	Flag unsafe configurations; recalibrate drifting twins.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	OpenAPS					
Reproducibility	Yes	No	Partial	**Metrics Reported**	RMSE, Clarke zones, TIR, runtime					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	0.5	1	1	1	1	1	0.5	0.5
Component Focus	Control	**Safety-Critical scenario**	Glycaemia	**System Type**	System-level					
Contribution of the paper	Tests artificial-pancreas configurations using digital twins.									
Summary	Uses personalized digital twins from OpenAPS traces to test artificial-pancreas configurations and reveal unsafe glucose-target behavior.									
Any Other Comments										

Study ID	SID34	**Paper Title**	Generating Qualifiable Avionics Software: An Experience Report							
Publication Year	2015	**Source**	IEEE	**Authors**	Andreas Wölf; Norbert Siegmund; Sven Apel; Harald Kosch; Johann Krautlager; Guillermo Weber-Urbina					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: DOD					
Standards Referenced	DO-178C, DO-333 , DO-330, DOD-STD-2167A									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Requirements/evolution/variability									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical			
Quantification method	Empirical metrics	**Assumptions**	Explicit assumptions; Bounds assumptions; Distributions assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other: Maintenance				
Technique Category	Design-time modeling, Architectural techniques, Assurance/validation	**Technique Name**	Embedded Database (EDB)							
Inputs	SysML model	**Outputs**	Avionics tests							
Mitigation Action	Regenerate models and requalify changed artifacts.									
Evidence Maturity Level	Industrial case, Deployed	**Evaluation Setting**	Case study , Real hardware							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Evolution scenarios					
Reproducibility	Yes	No	Partial	**Metrics Reported**	effort, variants, change counts, overhead					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	0.5	1	0.5	0.5	1	1	1	1	1	0.5
Component Focus	Database	**Safety-Critical scenario**	Avionics	**System Type**	Component-level					
Contribution of the paper	Generates qualifiable avionics software and evidence.									
Summary	Reports Airbus experience generating qualifiable avionics software/assets from SysML templates to reduce DO-178 qualification effort across variants.									
Any Other Comments										

Study ID	SID35	**Paper Title**	Dynamic platforms for uncertainty management in future automotive E/E architectures							
Publication Year	2017	**Source**	ACM	**Authors**	Philipp Mundhenk; Ghizlane Tibba; Licong Zhang; Felix Reimann; Debayan Roy; Samarjit Chakraborty					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: IEEE, OMG, AUTOSAR					
Standards Referenced	ISO 26262, IEEE 802.1 TSN , AUTOSAR, OMG, AUTOSAR									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other: Requirements driver				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Software/timing, Comms/network, Security/adversary, Environment, Sensor, ML/data-driven									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Formal			
Quantification method	Design-space exploration; Schedulability checks	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other: Maintenance				
Technique Category	Design-time modeling, Runtime monitoring, Architectural techniques, Assurance/validation	**Technique Name**	-							
Inputs	Architecture models	**Outputs**	Deployment evidence							
Mitigation Action	Monitor, report, update, reschedule, fail over.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Discussion							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	-					
Reproducibility	Yes	No	Partial	**Metrics Reported**	-					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	0.5	0.5	0.5	0	0	0	1	0.5
Component Focus	Platform	**Safety-Critical scenario**	Updates	**System Type**	System-level					
Contribution of the paper	Surveys uncertainty management for dynamic vehicles.									
Summary	Argues future automotive E/E platforms need uncertainty management through isolation, variant simulation, monitored updates, redundancy, and secure orchestration.									
Any Other Comments										

Study ID	SID36	**Paper Title**	Dynamic safety cases for through-life safety assurance							
Publication Year	2015	**Source**	IEEE	**Authors**	Ewen Denney; Ganesh Pai; Ibrahim Habli					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: ICAO SARP					
Standards Referenced	ICAO SARP									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, Human interaction, Comms/network									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Calibration	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other: Maintenance				
Technique Category	Runtime monitoring, Runtime assurance, Assurance/validation, Architectural techniques	**Technique Name**	Dynamic Safety Cases (DSCs)							
Inputs	Safety argument	**Outputs**	Safety updates							
Mitigation Action	Terminate flight; revise system and arguments.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Illustrative Example, Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	UAS missions					
Reproducibility	Yes	No	Partial	**Metrics Reported**	-					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	1	0.5	0.5	0	0.5	0.5	0.5	1
1Component Focus	Assurance	**Safety-Critical scenario**	UAS	**System Type**	System-level					
Contribution of the paper	Defines dynamic safety cases for operations.									
Summary	Proposes dynamic safety cases that update assurance confidence using runtime monitoring and trigger maintenance/system actions when assumptions degrade.									
Any Other Comments										

Study ID	SID37	**Paper Title**	An Analysis of ISO 26262: Machine Learning and Safety in Automotive Software							
Publication Year	2018	**Source**	Google Scholar	**Authors**	Rick Salay; Rodrigo Queiroz; Krzysztof Czarnecki					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO 26262, ISO/PAS 21448									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	ML/data-driven, Human interaction, Sensor, Environment, Software/timing, Security/adversary									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation		**Assumptions**	Explicit assumptions						
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Architectural techniques, Runtime assurance, Design-time modeling		**Technique Name**	-						
Inputs	ISO text		**Outputs**	ISO recommendations						
Mitigation Action	Fuse redundant decisions; enforce safety envelope.									
Evidence Maturity Level	Conceptual		**Evaluation Setting**	Discussion						
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	-					
Reproducibility	Yes	No	Partial	**Metrics Reported**	technique percentages					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	0.5	0.5	0.5	N/A	0.5	0.5	1
Component Focus	Perception	**Safety-Critical scenario**	ADAS	**System Type**	Component-level					
Contribution of the paper	Analyzes ISO 26262 gaps for ML.									
Summary	Analyzes ISO 26262's mismatch with ML and recommends ML-aware hazards, data requirements, redundancy, safety envelopes, and intent-based techniques.									
Any Other Comments										

Study ID	SID38	**Paper Title**	Assuring the Safety of Machine Learning for Pedestrian Detection at Crossings							
Publication Year	2020	**Source**	Springer	**Authors**	Lydia Gauerhof; Richard D. Hawkins; Chiara Picardi; Colin Paterson; Yuki Hagiwara; Ibrahim Habli					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: SAE					
Standards Referenced	ISO 26262, SAE J3016									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, ML/data-driven, Human interaction, Software/timing									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical			
Quantification method	Statistical estimation	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Design-time modeling	**Technique Name**	-							
Inputs	ODD/dataset	**Outputs**	Assurance gaps							
Mitigation Action	Restrict ODD, fail safe, augment data.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Experiment							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	JAAD					
Reproducibility	Yes	No	Partial	**Metrics Reported**	LAMR, miss rate, FPPI					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	0.5	1	0.5	0.5	1	0.5	1	0.5
Component Focus	Perception	**Safety-Critical scenario**	Crossing	**System Type**	Component-level					
Contribution of the paper	Derives ML safety requirements from scenarios.									
Summary	Decomposes pedestrian-crossing safety into ML perception/data requirements and shows JAAD/CNN evidence gaps that weaken the safety case.									
Any Other Comments										

Study ID	SID39	**Paper Title**	Safety assurance of Machine Learning for autonomous systems							
Publication Year	2025	**Source**	Science Direct	**Authors**	Colin Paterson; Richard Hawkins; Chiara Picardi; Yan Jia; Radu Calinescu; Ibrahim Habli					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: Def Stan, GSN					
Standards Referenced	ISO 21448 , ISO 26262, ISO/PAS 8800 , ISO/PAS 1883:2020 , Def Stan 00-56 Issue 4, GSN Standard v2									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Model/physics, Human interaction, ML/data-driven									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Formal			
Quantification method	Worst-case bounds	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Formal verification, Architectural techniques	**Technique Name**	AMLAS							
Inputs	Safety requirements	**Outputs**	Safety case							
Mitigation Action	Monitor assumptions; maintain evidence validity lifecycle.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Illustrative Example							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	ODD variability					
Reproducibility	Yes	No	Partial	**Metrics Reported**	mAP, IOU, robustness, FP/FN					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	1	1	0.5	0.5	0.5	1	0.5
Component Focus	Perception	**Safety-Critical scenario**	Stop-sign	**System Type**	Component-level					
Contribution of the paper	Introduces AMLAS process and argument patterns.									
Summary	AMLAS structures ML safety assurance from hazards to ML requirements, data requirements, model evidence, verification, and deployment assumptions.									
Any Other Comments										

Study ID	SID40	**Paper Title**	SMIRK: A machine learning-based pedestrian automatic emergency braking system with a complete safety case							
Publication Year	2022	**Source**	Science Direct	**Authors**	Kasper Socha; Markus Borg; Jens Henriksson					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: SAE					
Standards Referenced	ISO/PAS 21448:2019 , SAE J3016									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, ML/data-driven, Model/physics									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Ranking			
Quantification method	Calibration; Worst-case bounds	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Runtime monitoring, Architectural techniques, Assurance/validation	**Technique Name**	SMIRK							
Inputs	Sensor data	**Outputs**	Brake verdicts							
Mitigation Action	Reject OOD input; suppress brake signal.									
Evidence Maturity Level	Simulation	**Evaluation Setting**	Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	SMIRK images					
Reproducibility	Yes	No	Partial	**Metrics Reported**	false positives, false negatives					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	0.5	1	0.5	1	0.5	1	0.5
Component Focus	Perception	**Safety-Critical scenario**	PAEB	**System Type**	System-level					
Contribution of the paper	Publishes SMIRK demonstrator with safety case.									
Summary	SMIRK is an open PAEB demonstrator with radar/camera, YOLOv5, OOD safety cage, dataset, software, and safety-case artifacts.									
Any Other Comments										

Study ID	SID41	**Paper Title**	ThirdEye: Attention Maps for Safe Autonomous Driving Systems							
Publication Year	2022	**Source**	ACM	**Authors**	Andrea Stocco; Paulo J. Nunes; Marcelo d'Amorim; Paolo Tonella					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	ISO/PAS 21448:2019									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Environment, ML/data-driven, Software/timing, Model/physics, Sensor									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Statistical			
Quantification method	Calibration; Statistical estimation		**Assumptions**	Explicit assumptions; Bounds assumptions						
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Runtime monitoring, Runtime assurance, Assurance/validation		**Technique Name**	ThirdEye						
Inputs	Driving images		**Outputs**	Warning scores						
Mitigation Action	Trigger warning, disengage or request takeover.									
Evidence Maturity Level	Simulation		**Evaluation Setting**	Simulator/tool						
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	OOD simulations					
Reproducibility	Yes	No	Partial	**Metrics Reported**	precision, recall, F3, lead time, false alarms					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	0.5	1	0.5	1	1	1	1	0.5	0.5
Component Focus	Monitoring	**Safety-Critical scenario**	Lanekeeping	**System Type**	System-level					
Contribution of the paper	Predicts driving failures from attention maps.									
Summary	ThirdEye monitors lane-keeping DNNs using SmoothGrad attention maps and statistical thresholds to warn about off-road failures before they happen.									
Any Other Comments										

Study ID	SID42	**Paper Title**	Out-of-Distribution Detection as Support for Autonomous Driving Safety Lifecycle							
Publication Year	2023	**Source**	Springer	**Authors**	Jens Henriksson; Stig Ursing; Murat Erdogan; Fredrik Warg; Anders Thorsén; Johan Jaxing; Ola Örsmark; Mathias Örtenberg Toftås					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other: SAE					
Standards Referenced	ISO 26262 , ISO 21448 , ISO/TR 4804:2020, ISO/IEC AWI TR 5469, ISO/PAS 8800, SAE J3016									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Environment, Sensor, ML/data-driven, Model/physics, Security/adversary									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Statistical estimation; Calibration	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Runtime monitoring, Assurance/validation, Design-time modeling, Runtime assurance	**Technique Name**	-							
Inputs	Sensor streams	**Outputs**	OoD evidence							
Mitigation Action	Reduce ODD, stop, or transfer control.									
Evidence Maturity Level	Conceptual	**Evaluation Setting**	Illustrative Example, Case study							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Not Specified					
Reproducibility	Yes	No	Partial	**Metrics Reported**	FN rate, FPPI, OoD frequency, error margins					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	0.5	0.5	0	0.5	0.5	1	0.5
Component Focus	Perception	**Safety-Critical scenario**	AEB	**System Type**	Hybrid					
Contribution of the paper	Connects OoD detection to safety lifecycle.									
Summary	Positions OOD detection as a lifecycle bridge between ML metrics and ADS safety requirements, supporting ODD refinement and runtime fallback.									
Any Other Comments										

Study ID	SID43	**Paper Title**	Safety-Assured Formal Model-Driven Design of the Multifunction Vehicle Bus Controller							
Publication Year	2016	**Source**	Springer	**Authors**	Yu Jiang; Han Liu; Houbing Song; Hui Kong; Ming Gu; Jiaguang Sun; Lui Sha					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other:				
Standards Family	DO	ISO	IEC	EN	Other:					
Standards Referenced	IEC 61375, IEC 61375-1, IEC 61375-2									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Comms/network, Software/timing, Environment									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other: Formal			
Quantification method	Worst-case bounds	**Assumptions**	Explicit assumptions; Bounds assumptions							
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Formal verification, Design-time modeling, Runtime monitoring, Assurance/validation	**Technique Name**	-							
Inputs	IEC models	**Outputs**	Formal artifacts							
Mitigation Action	Flag monitor violations; no degradation specified.									
Evidence Maturity Level	Deployed, Industrial case	**Evaluation Setting**	Real hardware, Simulator/tool							
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Conformance tests					
Reproducibility	Yes	No	Partial	**Metrics Reported**	violations, binary size, bug detection					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	0.5	0.5	1	0.5	0.5	1	0.5
Component Focus	Communication	**Safety-Critical scenario**	Train	**System Type**	Component-level					
Contribution of the paper	Formalizes MVBC design from IEC standards.									
Summary	Formalizes IEC-61375 MVBC requirements in UPPAAL/TCTL, generates C, and adds runtime verification, uncovering standard issues and validating on rail hardware.									
Any Other Comments										

<table>
<tr><td>Study ID</td><td>SID44</td><td>Paper Title</td><td colspan="7">Integrating scenario- and contract-based verification for automated vessels</td></tr>
<tr><td>Publication Year</td><td>2024</td><td>Source</td><td>Springer</td><td>Authors</td><td colspan="5">Georg Hake; David Reiher; Jan Mentjes; Axel Hahn</td></tr>
<tr><td>Application Domain</td><td>Automotive</td><td>Aviation/UAV</td><td>Rail</td><td>Medical</td><td>Generic CPS</td><td colspan="4">Other: Maritime</td></tr>
<tr><td>Standards Family</td><td>DO</td><td>ISO</td><td>IEC</td><td>EN</td><td colspan="5">Other: ISO/IEC/IEEE</td></tr>
<tr><td>Standards Referenced</td><td colspan="9">ISO 24060:2021, ISO 26262, ISO 17894, ISO/IEC/IEEE 29148:2018</td></tr>
<tr><td>Role of Standard</td><td>Compliance</td><td>Guidance</td><td>Critique</td><td>Mapping</td><td>Terminology only</td><td colspan="4">Other:</td></tr>
<tr><td>Uncertainty Handled</td><td>Explicit</td><td colspan="2">Implicit</td><td>Type of Uncertainty</td><td>Aleatory</td><td colspan="2">Epistemic</td><td colspan="2">Both</td></tr>
<tr><td>Uncertainty Sources</td><td colspan="9">Sensor, Environment, Software/timing, Model/physics</td></tr>
<tr><td>What is Uncertain?</td><td>Variables</td><td>Parameters</td><td>Assumption</td><td colspan="6">Other:</td></tr>
<tr><td>Representation</td><td>Probabilistic</td><td>Intervals</td><td>Sets</td><td>Scenarios</td><td>Evidence</td><td>Qualitative</td><td colspan="3">Other:</td></tr>
<tr><td>Quantification method</td><td colspan="3">Worst-case bounds</td><td>Assumptions</td><td colspan="5">Explicit assumptions; Bounds assumptions</td></tr>
<tr><td>Lifecycle Phases</td><td>Requirements</td><td>Design</td><td>Implementation</td><td>V&V</td><td>Runtime/ operations</td><td colspan="4">Other: Maintenance</td></tr>
<tr><td>Technique Category</td><td colspan="3">Formal verification, Assurance/validation, Runtime monitoring</td><td>Technique Name</td><td colspan="5">Verification Descriptors (VDs)</td></tr>
<tr><td>Inputs</td><td colspan="3">Requirements/scenarios</td><td>Outputs</td><td colspan="5">Contract monitors</td></tr>
<tr><td>Mitigation Action</td><td colspan="9">Warn, revise implementation, reject breached update.</td></tr>
<tr><td>Evidence Maturity Level</td><td colspan="3">Industrial case, Lab-testbed</td><td>Evaluation Setting</td><td colspan="5">Simulator/tool</td></tr>
<tr><td>Baselines/ Comparators</td><td>Yes</td><td>No</td><td>Partial</td><td>Dataset / Scenarios</td><td colspan="5">Scenario catalog</td></tr>
<tr><td>Reproducibility</td><td>Yes</td><td>No</td><td>Partial</td><td>Metrics Reported</td><td colspan="5">contract breaches, deviation, latency</td></tr>
<tr><td rowspan="2">Applied QARs</td><td>Q1</td><td>Q2</td><td>Q3</td><td>Q4</td><td>Q5</td><td>Q6</td><td>Q7</td><td>Q8</td><td>Q9</td></tr>
<tr><td>1</td><td>1</td><td>0.5</td><td>1</td><td>1</td><td>0.5</td><td>1</td><td>0.5</td><td>1</td></tr>
<tr><td>Component Focus</td><td colspan="2">Monitoring</td><td>Safety-Critical scenario</td><td colspan="2">Berthing</td><td>System Type</td><td colspan="3">Component-level</td></tr>
<tr><td>Contribution of the paper</td><td colspan="9">Unifies scenario and contract-based vessel verification.</td></tr>
<tr><td>Summary</td><td colspan="9">Integrates scenario-based and contract-based verification for automated vessels via a Verification Descriptor and monitorable contracts for update gating.</td></tr>
<tr><td>Any Other Comments</td><td colspan="9"></td></tr>
</table>

Applied QARs (cont.)	Q10
	0.5

Study ID	SID45	**Paper Title**	Towards dynamic safety assurance for Industry 4.0							
Publication Year	2021	**Source**	Science Direct	**Authors**	Muhammad Atif Javed; Faiz Ul Muram; Hans Hansson; Sasikumar Punnekkat; Henrik Thane					
Application Domain	Automotive	Aviation/UAV	Rail	Medical	Generic CPS	Other: Industrial				
Standards Family	DO	ISO	IEC	EN	Other: GSN, OMG, ANSI					
Standards Referenced	GSN Standard v2 , OMG, ANSI/ITSDF B56.5, IEC 61508, ISO 13849-1, IEC 61496-1									
Role of Standard	Compliance	Guidance	Critique	Mapping	Terminology only	Other:				
Uncertainty Handled	Explicit	Implicit	**Type of Uncertainty**	Aleatory	Epistemic	Both				
Uncertainty Sources	Sensor, Environment, Software/timing, Comms/network, Human interaction									
What is Uncertain?	Variables	Parameters	Assumption	Other:						
Representation	Probabilistic	Intervals	Sets	Scenarios	Evidence	Qualitative	Other:			
Quantification method	Worst-case bounds			**Assumptions**	Explicit assumptions; Bounds assumptions; Independence assumptions					
Lifecycle Phases	Requirements	Design	Implementation	V&V	Runtime/ operations	Other:				
Technique Category	Assurance/validation, Runtime monitoring, Architectural techniques, Design-time modeling			**Technique Name**	-					
Inputs	Safety artifacts			**Outputs**	Safety updates					
Mitigation Action	Degrade, reconfigure sensing, update safety case.									
Evidence Maturity Level	Simulation , Conceptual			**Evaluation Setting**	Case study , Simulator/tool					
Baselines/ Comparators	Yes	No	Partial	**Dataset / Scenarios**	Smart factory					
Reproducibility	Yes	No	Partial	**Metrics Reported**	requirement satisfaction, safety outcomes					
Applied QARs	**Q1**	**Q2**	**Q3**	**Q4**	**Q5**	**Q6**	**Q7**	**Q8**	**Q9**	**Q10**
	1	1	0.5	1	0.5	0.5	0.5	0.5	1	0.5
Component Focus	Assurance	**Safety-Critical scenario**	AGV	**System Type**	System-level					
Contribution of the paper	Links safety contracts to dynamic safety cases.									
Summary	Links Industry 4.0 safety cases to runtime safety contracts so operational deviations can update both contracts and the safety argument.									
Any Other Comments										